

Report

Of

Ambrosia - A B2B Canteen & Food Ordering Platform for Industrial Workforces

Submitted To

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B.Tech CSE

By

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CANDIDATE'S DECLARATION

I, Raghvendra Kumar, do hereby declare that the internship report titled "**Ambrosia - A B2B Canteen & Food Ordering Platform for Industrial Workforces**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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4. PROJECT DESCRIPTION

4.1 Introduction

Ambrosia is a B2B food ordering and canteen management application developed during my internship at Lobra Hansetu Private Limited (Hansetu), as part of the company's industrial platform offerings for business clients. The module is aimed at industrial units and factory campuses that need a digital way to manage canteen ordering for their workforce letting employees browse the day's menu, place food orders, and track order status, while giving the client's canteen/vendor team a simple system to manage orders in real time. The application is built using Flutter for the front end, allowing a single codebase to run on both Android and iOS, and Firebase for backend services such as authentication, database, and storage.

4.2 Organization Profile

I completed this internship at Lobra Hansetu Private Limited (Hansetu), a company working on a B2B industrial platform as well as mobile application development for business clients. The organization's key details are as follows:

Company Name	Lobra Hansetu Private Limited (Hansetu)
CIN	U62099HR2024PTC123174
Registered Office	SCO 8FF, 1st Floor, Plot No 658/685, Anand Market, Buta Singh Colony, Meera Ghati Chowk, Karnal, Haryana - 132001
Website	www.hansetu.com
Nature of Business	B2B industrial platform and mobile application development
My Role	Developer Team Member
Duration	26/05/2026 to 25/07/2026

4.3 Problem Statement

Industrial units and factory campuses typically manage staff canteen operations manually through paper tokens, verbal orders, or informal coordination which leads to long queues, order errors, and no visibility into daily demand for the canteen/vendor team. As part of Hansetu's B2B industrial platform, there was a need for a lightweight ordering module that industrial clients could use to digitize canteen ordering for their workforce without adding unnecessary complexity.

4.4 Project Objectives

- To design and develop a cross-platform B2B canteen ordering application using Flutter.
- To integrate a real-time backend using Firebase for authentication and data management.

- To implement a complete ordering flow for industrial staff: browsing the menu, cart, and order placement.
- To give the canteen/vendor side real-time visibility into incoming orders through the same Firebase backend.
- To prepare the application for a production-ready release, including UI polish and a signed release build.

4.5 Scope of the Project

The scope of the project covers the design and development of the mobile application front end in Flutter, integration with Firebase for authentication and Firestore for order data storage, and preparation of the app for a release build, aimed at industrial/B2B clients of Hansetu's platform. The project also explores further production-readiness features such as location search and payment gateway integration, which are planned as future enhancements beyond the current build.

4.6 Technologies and Tools Used

Layer	Technology
Frontend/App Framework	Flutter (Dart)
Backend Services	Firebase Authentication, Firestore, Firebase Storage
Planned Integrations	Google Places API, Razorpay Payment Gateway
Version Control	Git & GitHub

4.7 System Architecture

The application follows a client-backend architecture suited to a B2B deployment. The Flutter application acts as the client layer, handling the UI and user interactions across screens such as the menu/canteen listing, cart, and order tracking, used by employees of the industrial client. Firebase serves as the backend, with Firebase Authentication managing user sign-in, Cloud Firestore storing menu, order, and vendor data in real time so the canteen/vendor team can see incoming orders as they come in, and Firebase Storage handling image assets such as menu photos and the app icon. The Flutter client communicates directly with Firebase services through the official Firebase SDKs, keeping the architecture simple and serverless, without a separate custom backend server which also keeps it easy to deploy across multiple industrial client sites.

4.8 Methodology

1. **Requirement analysis:** identifying the core features needed for a B2B canteen ordering module.
2. **UI/UX design:** designing the app screens and user flow for browsing, cart, and checkout.
3. **Flutter app development:** building the core screens and navigation.
4. **Firebase integration:** setting up authentication, Firestore collections, and storage.

5. **Testing:** verifying the ordering flow end to end on test data.
6. **Release preparation:** UI theme finalization, real data seeding, app icon, and building the release APK.

4.9 Expected Outcomes

- A working, functional B2B canteen ordering mobile application with a complete ordering flow.
- Hands-on experience with Flutter for cross-platform mobile development.
- Practical understanding of integrating a serverless backend (Firebase) into a mobile app.
- A release-ready APK build published to GitHub, deployable across industrial client sites.

4.9.1 Application Screens

The following screens illustrate the key parts of the application's user interface.

Fig 4.9.1(a): Home Screen - browsing the day's canteen menu

Fig 4.9.1(b): Cart Screen - order summary and checkout

Fig 4.9.1(c): Order Tracking Screen

4.10 Certificates and Communication Proof

The internship completion certificate issued by the organization is attached at the front of this report (Chapter 3). Any additional communication/mail proof from the organization or mentor can be attached following this section.

5. BIBLIOGRAPHY/REFERENCES

- **Flutter Documentation:** <https://docs.flutter.dev>
- **Firebase Documentation:** <https://firebase.google.com/docs>
- **Dart Programming Language Documentation:** <https://dart.dev/guides>